B777/B787 Pacific Reference Guide

Flight Planning

- ❑ Download EFLIGHT PLAN (CCI)
- ❑ Evaluate Track Message:
  - check date/time for validity
  - check remarks
  - if filed on published track, compare track waypoints to Oceanic Route Verification section of flight plan
- ❑ Crew passports/visas
- ❑ Review Chart Change Notices/NOTAMS
- ❑ AIREP Form (MISC 84)
- ❑ Download WSI Weather Charts:
  - Wind/Temp Forecast 250mb (FL 340)
  - High Level Sig Wx
- ❑ Volcanic Ash Info
- ❑ Check B777/B787 Fleet Bulletin 26-002: Oceanic Procedures for any updates

Preflight

- ❑ Fit for Duty
- ❑ General Declarations (if required).
- ❑ Customs declarations forms (if required)
- ❑ ETOPS Pre-Departure Check .....Entered in AML
- ❑ Review AML for ETOPS inflight verification check (see Approaching Oceanic Entry Point on page 2).
- ❑ Water/LAV Service .....Verify accomplished
- ❑ Security Search.....Verify accomplished
- ❑ B777 only, SEL 28 (AOM 20.1.1)
  - Aircraft nose numbers: 7AA - 7CA *only*
  - 700 pounds of additional fuel *must* appear on the flight plan when RLS FUEL is greater than 117,000 pounds.
- ❑ NOTOC Acknowledged (if applicable)
- ❑ Verify iPad EFBs current including flight plan
- ❑ Lavatory PA: flights inbound to the US, 'We ask you not to congregate in groups in any area of the cabin especially around the lavatories. Thank you for your cooperation.'"

Enroute

- ❑ Change ICAO STEP on VNAV CRZ page to 1000 feet when evaluating cruise altitudes at OEP (FP, highest acceptable, OPT/MAX/RECMD).
- ❑ Update winds
- ❑ **Logon CPDLC** – when approaching area with CPDLC (10-25 minutes prior to entry).

ACARS Area Weather

ANC	PANC	Anchorage	SYD	YSSY	All Australian Airports
FAI	PAFA	Fairbanks	PKC	UHPP	Petropavlosk
LAX	KLAX	Los Angeles	PVG	ZSPD	Shanghai
NRT	RJAA	Narita/Haneda	SEA	KSEA	Seattle
PEK	ZBAA	Beijing	SFO	KSFO	San Francisco
AKL	NZAA	All New Zealand Airports	YEG	CYEG	Edmonton
HNL	PHNL	All Hawaiian Airports	SPAM	SPAM	South Pacific Alternate Airports

CPDLC Logon (Jeppesen Airway Manual)

BIRD	Reykjavik	RJJJ	Fukuoka
CDQX	Gander Domestic	SCEZ	Isla De Pascua
CZEG	Edmonton	VHHH	Hong Kong
CZQM	Moncton	VIDF	Delhi
CZQX	Gander Oceanic	VVTS	Ho Chi Minh
CZUL	Montreal	WAAF	Ujung Pandang
CZWG	Winnipeg	WSJC	Singapore
CZVR	Vancouver	YBBB	Brisbane, Honiara, Nauru
CZYZ	Toronto	YMMM	Melbourne
GDXB	Magadan	ZBAB	Beijing
KZAK	Oakland Oceanic	ZLLL	Lanzhou
NFFF	Nadi	ZMUA	Ulan Bataar
NTTT	Tahiti	ZPPP	Kunming
NZZO	Auckland Oceanic	ZULS	Lhasa
PAZA	Anchorage - DOM Arctic (north of N63 & east of W165)	ZUUU	Chenghu
PAZN	Anchorage - Oceanic (south of N63 & west of W165)	ZWWW	Urum-qi
RKTT	Seoul	ZYHB	Harbin

SATCOM Quick Reference

May not work above 82° North.
Obtain SFO SELCAL before 80° North.

Note: For ATC communications CPDLC may be more effective.

SATCOM INMARSAT (B777)

1. MENU, SAT (2L), AA DISP #s [5R].
2. Select Dispatch Desk #.
3. For INMARSAT select ATC emergency from SAT MAIN MENU scratch pad, enter in 1L or 3L.
4. PLACE CALL (2R or 4R).

SATCOM INMARSAT (B787)

1. Select SAT on the TCP.
2. From the SAT PHONE page, if the desired dispatch desk (DSP DSK XX) or ground station is displayed in line 1L or 3L, proceed to step 4.
3. To select a ground station from the SAT DIRECTORY, select DIRECTORY at line 2R. Select the desired ground station to SAT 1 or SAT 2. The display will return to the SAT PHONE page.
4. If required, select a priority (EMG, HGH or LOW). Press the MAKE CALL for the desired SAT radio.

Cost Index/ECON Speed Procedures

While operating in Oakland (KZAK) oceanic control area, in order to support cost index/ECON speeds and maintain ATC separation, inform ATS via VOICE or CPDLC each time the cruising MACH number varies or is expected to vary by a value equal to or greater than 0.02 MACH from:

— the MACH number at the FIR entry point or

— any subsequent speed change notified to ATC in flight

Flight Information Region (FIR)

CPDLC Position Report (Pacific FIRs)

All aircraft with an active ADS connection should make one CPDLC position report over the FIR boundary (exception: KZAK and RJJJ).

SFO Radio

- ❑ Contact SFO Radio for a HF SELCAL check before entering OAK FIR (SFO only - state HF freq).
- ❑ When not in VHF contact with Anchorage Center, contact SFO Radio on HF (e.g., westbound).

Example: "San Francisco Radio, American ### on (HF freq), SELCAL check." Expect current primary and secondary along with transition frequencies to be used crossing 140 degrees west.

Russia

- ❑ Report each compulsory fix:
  - position/time/flight level
  - next compulsory fix/ETA

Note: Expect request for FIR exit fix time estimate.

See JeppFD-Pro for all current HF/VHF frequency listings.

Oceanic Route Check

If ATC issues a route amendment:

- ❑ PF and PM will separately review the amended clearance vs FMC
- ❑ PM will compare amended clearance to FMC LEGS (expand all latitude/longitude waypoints) and note any changes
- ❑ PF will compare amended clearance to FMC LEGS (expand all latitude/longitude waypoints); any differences will be immediately changed in the FMC by the PF while the PM confirms the accuracy of the changes
- ❑ Advise dispatch of revised routing
- ❑ Obtain new Critical Fuel Scenario and Oceanic Route Verification
- ❑ Obtain new FLT PROGRESS/FPR
- ❑ One pilot (PM or PF) will read aloud FMC waypoints while the other pilot checks them against the amended clearance
- ❑ Do not respond to CPDLC CONFIRM ASSIGNED ROUTE message until all route amendments are entered in the FMC, confirmed, and executed
- ❑ Update winds for new route/altitude

If no route amendment is issued by ATC:

- No further legs check is required if a successful preflight route and legs check has been accomplished

Note: Half-degree coordinates can only be checked by expanding the waypoints. Caution is required to ensure these coordinates are correctly entered in the FMS.

Prior to Oceanic Entry Point (OEP)

- ❑ If required by a TAC item, determine the ETOPS entry/exit point using the fix page or lat/long from the Critical Fuel Scenario.

ETOPS Alternate Planning/Diversion Guide:

- ❑ Update alternate WX, forecast, and NOTAMs.
- ❑ Evaluate potential diversion airport(s): **FIRE**, Mechanical, Medical/Passenger.

Approaching Oceanic Entry Point

- ❑ Check ECON speed (cost index) or assigned MACH
- ❑ Confirm at cleared cruise altitude and set in FMC
- ❑ If required, complete any **ETOPS In Flight Verification Check(s)**; such as an in flight engine check, an in flight APU start, etc., prior to ETOPS entry.

Note: Ensure the required AML entries are made prior to entering ETOPS airspace.

After Oceanic Entry Point

- ❑ Operational HF/SELCAL check (if not already done).
- ❑ Tune left VHF radio to 121.5 and right VHF to 123.45.
- ❑ Transponder: As directed by ATC. Absent any ATC instructions, leave transponder on current code. *Exception*: Westbound into RJJJ, select 2000 crossing 164E longitude.
- ❑ Establish SLOP [see Strategic Lateral Offset Procedure (SLOP) for more information].

Note: SLOP is required in all oceanic airspace.

Oceanic Procedures
PACOTS and Random Routes Only

Approaching Active Oceanic Waypoint

Both pilots must verify that the expanded full Lat/Long coordinates for the next and the following waypoint agree with the active flight plan and/or ATC clearance, as well as the Oceanic Route Verification.

Waypoint Passage

1. Confirm next waypoint becomes active waypoint.
2. Verify autopilot is coupled to LNAV.
3. Record time and fuel on FPR.
4. Complete AIREP Form/transmit report (if required).
5. Evaluate potential diversion airport(s): fire, mechanical, medical/passenger

Midpoint Check

1. Check ETA for active waypoint. If ETA has changed by 3 minutes or more from what was reported to ATC revise estimate using CPDLC, ATC free text, or voice. If transmitting position via ADS-C, no need to update ETA.
2. Check for satisfactory fuel quantity/balance and trend.

Oceanic Exit

1. Confirm ATC clearance (if required).
2. If applicable, communicate acceptance/rejection of redispatch message. Redispatch may be received up to 2 hours prior; however, if not received within 1 hour from redispatch point, contact the dispatcher.
3. Request desired altitude and Mach from ATC.
4. Confirm SLOP offset is zero.

   Note: Do not go direct to the oceanic exit point.
5. Change STEP on VNAV CRZ page to 2000 feet to evaluate cruise altitudes after oceanic exit (FP, highest acceptable, OPT/MAX/RECMD)

Standard Entry/Exit Briefing

Prior to PF or captain leaving the flightdeck or after any assigned pilot re-enters the flightdeck while enroute:

- ❑ PF brief/PM review aircraft status from FMA.
- ❑ PF brief additional items (aircraft systems status, weather, ATC clearance/frequency change).
- ❑ For a crew rest break the captain/PIC will brief:
  - a designated PIC
  - weather deviation options and procedures
  - current and *expect* clearances (e.g., redispatch, step climbs, etc.)
  - diversion options to include alternate airports and weather, departure from track/airway procedures, ETPs, and terrain awareness areas
  - any other operational considerations (e.g., datalink clearances, required radio frequency or transponder code changes)
  - crew rest location(s) (bunk or rest seat) as applicable

Strategic Lateral Offset Procedure(SLOP)

SLOP is a standard operating procedure for the entire region and applies for organized tracks (OTS) and random routes. The options are a 1 or 2 nm offset to the right, or centerline.

Note: Aircraft (e.g., B787) able to perform offsets in tenths of nautical mile should do so as it contributes to risk reduction.

1. Offset using LNAV, not heading select.
2. Do not advise ATC or request ATC clearance.

Return to course by oceanic exit point by deleting offset.

Set Max Uplink Delay

1. Select ATC from Communication Main Menu.
2. Select LOGON/STATUS.
3. Set MAX UPLINK DELAY to 300 seconds or requested value.

Confirm Assigned Route

- ❑ Select DISPLAY REPORT prompt.
- ❑ ATC REPORT page shows SLOP as DEV.
- ❑ Select SEND.

AIREP Form (FO-0024)

Use the AIREP form when:

— operating in (either the North Atlantic or Pacific oceanic airspace) and non-radar position reporting is required

— in other areas as desired

Note: Available via ACARS - MISC 84

Loss of Cabin Pressure

See AOM 17.6 Emergency Descent. This procedure is in ECL> Unannunciated Checklists and QRH> Miscellaneous. Turn toward a suitable airfield when appropriate.

Medical

1. Refer to FOM> Fast Reference Links> Medical Emergencies.
2. If Big 4, diversion required: notify Captain, Dispatch. Obtain revised clearance. If unable, accomplish Departure from Route/Track on page 5

Weather Deviations

Weather Deviations

Revised Clearance

If a lateral deviation from a route/track is necessary to avoid weather (e.g., thunderstorms), request a revised ATC clearance as soon as possible by stating “Weather deviation required.” or by a CPDLC lateral down link message.

1. Select “COMM MENU”
2. Select “ATC”
3. Select “ROUTE REQUEST”, then “WX DEVIATION”
4. Enter requested deviation mileage and direction. If no direction is entered, controller will assume requested deviation is left and right of course.

Obtain essential traffic information prior to deviating; even for a small distance (e.g., less than 5 nm). If a revised ATC clearance cannot be obtained, follow the procedures described below.

After the deviation is started, continue to obtain a revised ATC clearance.

Deviation Beyond ATC Clearance Parameters

ATC can only provide a revised clearance which will continue to assure minimum standard separations are maintained from all other traffic, including any adjacent routes/tracks.

When this is not possible, ATC will advise “Unable and state intentions.”

The pilot should indicate the direction, anticipated magnitude and, if appropriate, the expected altitude change of the intended deviation.

Deviation with ATC Clearance

If a revised clearance is received, it should be followed in all its provisions.

If such a revised clearance is received after beginning the contingency procedures, the revised clearance supersedes the contingency procedure provisions.

Deviation without ATC Clearance

If revised ATC clearance has not been obtained, use the following procedures:

1. Deviate away from the organized track or route system, if possible.
2. Establish communications with and alert nearby aircraft broadcasting, at suitable intervals:
   - aircraft identification, flight level, aircraft position (including ATS route designator or the track code) and intentions.

   Broadcast on the frequency in use (when VHF) and on frequency 121.5 MHz (or, as a back-up, on the VHF inter-pilot air-to-air frequency 123.45 MHz) or use CPDLC.

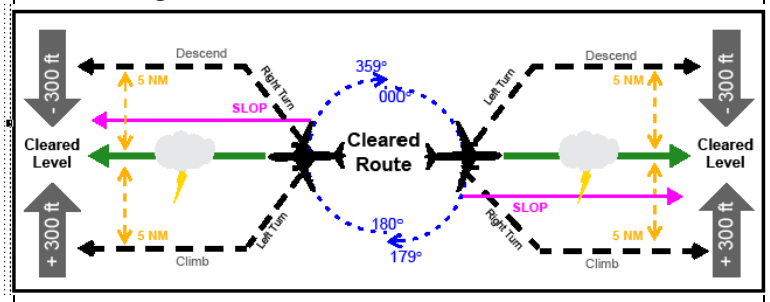
   Note: When necessary use the urgency call “pan-pan” or the CPDLC urgency down link message.
3. Watch for conflicting traffic both visually and by reference to TCAS.
4. Turn on all aircraft exterior lights.
5. For deviations of:
   - **less than 5** nautical miles:
     remain at the ATC assigned flight level
   - **equal to or greater than 5** nautical miles:
     when the aircraft is 5 nautical miles or greater from the magnetic track/route centerline, initiate a level change of 300 feet

   Note: Adjust the aircraft path to avoid conflict with other aircraft at or near the same flight level.

   When the magnetic track/route centerline is:

   — east 000° - 179° and you are:
   - **left** of course: descend 300 feet
   - **right** of course: climb 300 feet

   — west 180° - 359° and you are:
   - **left** of course: climb 300 feet
   - **right** of course: descend 300 feet



Example: Your magnetic track/ route centerline is 175° and you are deviating 5 nm (and possibly more) to the right of your course: you need to initiate a 300 foot climb when passing 5 nm.

When returning to route/track:

6. Resume the last assigned flight level when the aircraft is less than 5 nm from the centerline.

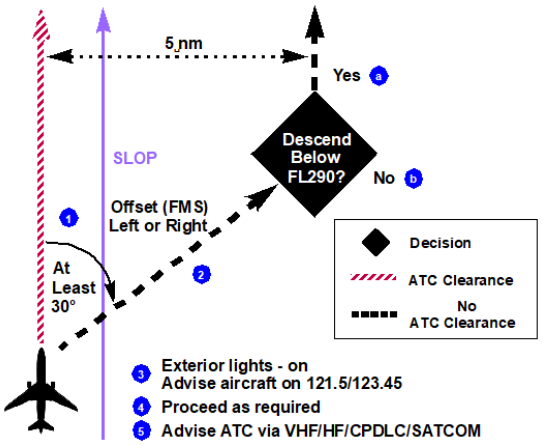
Deviation Complete

Advise ATC when the weather deviation is no longer required, or when a weather deviation has been completed and the aircraft has returned to the centerline (or previously adopted SLOP offset) of its cleared route.

Note: Weather deviations beyond 128 nm are not possible due to FANS limitations. If more than 128 nm is needed, the flight crew must submit a Route Request containing a waypoint that will avoid the weather. Include in the Route Request a downtrack waypoint on the flight plan, past the weather.

Departure from Route/Track

Departure from Route/Track



If unable to continue flight in accordance with an ATC clearance due to inflight contingencies (equipment failure, etc.) or an emergency, obtain a revised clearance, whenever possible, prior to initiating action. Contact ATS via CPDLC/VHF/HF/SATCOM, or request dispatch via SATCOM to contact ATS, to advise of actions and request clearance (time and conditions permitting).

Unable to Obtain Clearance

If unable to obtain an ATC clearance in a timely manner, the captain should exercise emergency authority by executing whatever action necessary to ensure flight safety.

Note: The aircraft should be flown at a flight level and altitude where other aircraft are less likely to be encountered.

Use the following procedures (also located on the specific-region reference guides) until a revised clearance is received:

1. Leave the assigned route or track by turning at least 30° to the left or right (offset/LNAV) based on:
   - aircraft position relative to any organized track or route system,
   - direction of flights and flight levels allocated on adjacent tracks,
   - direction to an alternate airport,
   - any strategic lateral offset being flown, and
   - terrain clearance.
2. Maintain the assigned flight level (if possible) until established on the 5 nm parallel, same direction track/route offset.

   Note: If unable to maintain the assigned flight level, minimize climb/descent rate to the extent that is operationally feasible.

   Once established on a 5 nm parallel, same direction track/route offset, decide whether to continue descent to below FL290 (**recommended**).

   Descent below FL290 is particularly applicable to operations where there is a predominant traffic flow (e.g., east-west) or parallel track system where the aircraft's diversion path will likely cross adjacent tracks or routes. Descending below FL290 can decrease the likelihood of conflicts with other aircraft, TCAS RA events, and delays in obtaining a revised ATC clearance.

   a. If continuing the descent to below FL290 and once below FL290:

      establish/maintain an altitude that differs by 500 feet from altitudes normally used.

   b. If <u>not</u> continuing the descent:

      establish/maintain an altitude that differs by 500 feet (1000 feet if above FL410) from altitudes normally used.

   Note: Altimetry system error may lead to less than actual 500 feet vertical separation and TCAS RAs may occur.
3. Turn on all exterior aircraft lights (IAW aircraft operating manual limitations) and broadcast position and intentions at suitable intervals on appropriate air-to-air, air-to-ground frequencies and 121.5 MHz to alert other aircraft in the area using “pan-pan” (urgent) or “mayday” (distress) as appropriate. Proceed as required by the operational situation or if an ATC clearance has been obtained, proceed in accordance with the clearance.
4. Advise ATC: “mayday” or “pan-pan” (as appropriate) using CPDLC, SATCOM, or 121.5.
5. Record/annotate time and approximate position for any reports that may be required.
6. Submit appropriate pilot debrief (if necessary).

Russia Departing Track Procedures

Change of Flight Level

When encountering hazards to flight safety at the assigned altitude (flight level) (dangerous meteorological phenomena, failures of equipment, etc.), the pilot shall have the right to change the altitude (flight level) independently, with immediate reporting to the appropriate ATC unit.

In this case, the pilot shall, without changing the altitude (flight level), turn away, to the right 30 degrees from the airway axis and, after 16.2 nm (30 km), resume the former flight heading with simultaneous change of altitude to the selected level and report to the ATC unit. Maintain a 5 nm offset.

In an emergency, the descent shall be carried out immediately after turning way. Upon reaching the new flight level, the pilot shall, after receiving clearance from the ATC unit, enter the airway.

ICAO Cold Temperature Error Table

Height Above Airport in Feet

Record Temp°C	200	300	400	500	600	700	800	900	1000	1500	2000	3000	4000	5000
+10	10	10	10	10	20	20	20	20	20	30	40	60	80	90
0	20	20	30	30	40	40	50	50	60	90	120	170	230	280
-10	20	30	40	50	60	70	80	90	100	150	200	290	390	490
-20	30	50	60	70	90	100	120	130	140	210	280	420	570	710
-30	40	60	80	100	120	140	150	170	190	280	380	570	760	950
-40	50	80	100	120	150	170	190	220	240	360	480	720	970	1210
-50	60	90	120	150	180	210	240	270	300	450	590	890	1190	1500

Example: Temperature -10° C and the aircraft altitude is 1,000 feet above the airport elevation. The chart shows the reported current altimeter setting may place the aircraft as much as 100 feet **below** the altitude indicated by the altimeter.

Russian Airports and QFE Conversions

QFE (mb or hPA) + ΔP = QNH

Airport Name	ICAO	ATA	Elevation (ft)	Inches Hg	hPa	ΔP (hpa)
Anadyr	UHMA	DYR	194	0.194	6.570	7
Bratsk	UIBB	BTK	1610	1.61	54.521	55
Chita	UIAA	HTA	2270	2.27	76.871	77
Irkutsk	UIII	IKT	1675	1.675	56.722	57
Khabarovsk	UHHH	KHV	244	0.244	8.263	9
Krasnoyarsk	UNKL	KJA	942	0.942	31.900	32
Magadan	UHMM	GDX	574	0.574	19.438	20
Mirny	UERR	MJZ	1162	1.162	39.350	40
Murmansk	ULMM	MMK	266	0.266	9.008	9
Norilsk	UOOO	NSK	594	0.594	20.115	20
Novosibirsk	UNNT	OVB	365	0.365	12.360	13
Petropavlovsk	UHPP	PKC	131	0.131	4.436	5
Polinary	UERP	PYJ	1670	1.67	56.553	57
Ulan-Ude	UIUU	UUD	1693	1.693	57.332	58
Vladivostok	UHWW	VVO	46	0.046	1.558	2
Yakutsk	UEEE	YKS	325	0.325	11.006	11

Only For Use in Chinese, Mongolian, or North Korean Airspace

Meters to Feet Conversion			
Meters (RVSM bold)	Feet (RVSM bold)	Meters	Feet
13100	**43000**	5400	17700
12500	**41100**	5100	16700
12200	**40100**	4800	15700
11900	**39100**	4500	14800
11600	**38100**	4200	13800
11300	**37100**	3900	12800
11000	**36100**	3600	11800
10700	**35100**	3300	10800
10400	**34100**	3000	9800
10100	**33100**	2700	8900
9800	**32100**	2400	7900
9500	**31100**	2100	6900
9200	**30100**	1800	5900
8900	**29100**	1700	5600
8400	27600	1600	5200
8100	26600	1500	4900
7800	25600	1400	4600
7500	24600	1300	4300
7200	23600	1200	3900
6900	22600	1100	3600
6600	21700	1000	3300
6300	20700	900	3000
6000	19700	600	2000
5700	18700	300	1000

Russia Airspace

— Above transition level/transition altitude Russian airspace uses flight levels in feet.
— Below TL/TA use alt/height conversion charts on applicable STAR/SID/Approach Charts.
— Request QNH for conversion from meters to feet.
— Do not use QFE.
— Do not use meters switch on EFIS Control Panel.
— Do not use the above conversion chart in Russian airspace.

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